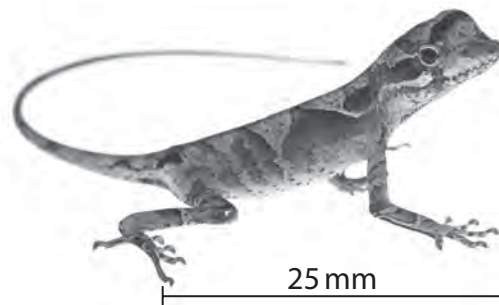


3 The photograph shows a lizard native to a Caribbean island near the equator.



© Nature Picture Library / Alamy Stock Photo

This small lizard lives in tropical vegetation between 1 m to 3 m above the ground.

This lizard uses specialised toepads to hold on to the vegetation.

A scientist sampled the lizards along a 2 km transect. The toepad area of a front limb of each lizard was measured.

The lizards were then released.

A hurricane struck the island with the highest winds ever recorded.

The scientist repeated the sampling method along the same transect after the hurricane.

The results of this investigation are shown below.

Surface area in mm² of toepad before the hurricane, sample **A**

0.9 1.3 1.1 1.2 1.3 1.5 1.6 0.8 0.7

Surface area in mm² of toepad after the hurricane, sample **B**

0.8 1.3 1.5 1.8 1.3 1.5 1.6 1.9 1.8

(a) State a suitable null hypothesis for this investigation.

(1)

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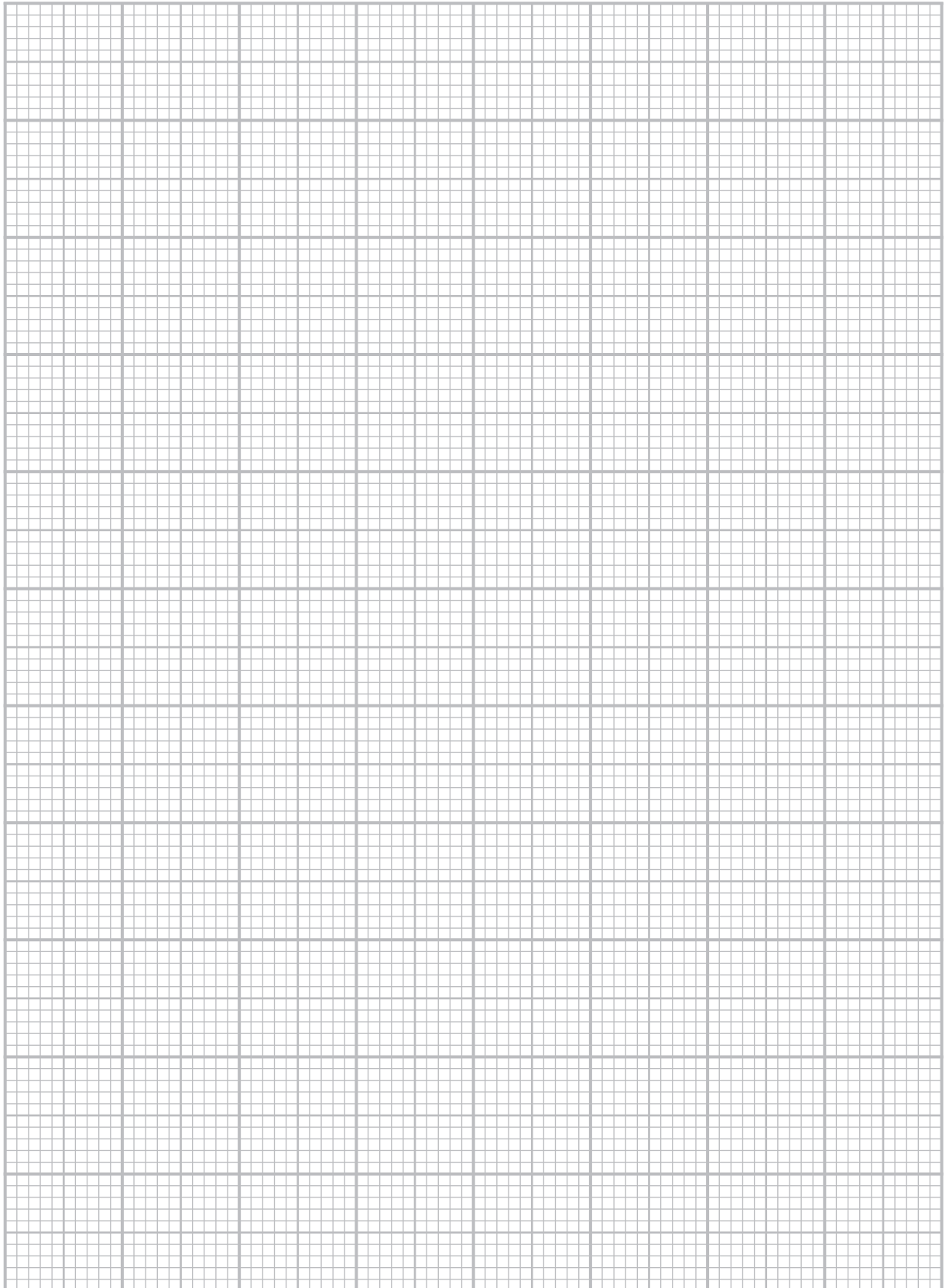
(b) Draw a suitable table to display the **data** and your calculated **means** for the toepad area of sample **A** and the toepad area of sample **B**.

(3)

- (c) Draw a suitable graph to show the mean toepad area for sample **A** and for sample **B**.

Include an indication of the variability of the data.

(3)



(d) The scientist analysed the data with a t test using the formula:

$$t = \frac{(\bar{x}_A - \bar{x}_B)}{\sqrt{\frac{(S_A)^2}{n_A} + \frac{(S_B)^2}{n_B}}}$$

Where:

$\bar{x}$ is the mean value for each treatment

n is the number of samples for each treatment

$(S_A)^2 = 0.095$ and $(S_B)^2 = 0.115$

(i) Calculate the value of t .

(3)

Answer

(ii) The table shows the critical values of t for different degrees of freedom.

The number of degrees of freedom = $(n_A - 1) + (n_B - 1)$

Degrees of freedom	$p = 0.05$
8	2.31
9	2.26
10	2.23
11	2.20
12	2.18
13	2.16
14	2.15
15	2.13
16	2.12
17	2.11
18	2.10

Describe the conclusions that can be drawn from this investigation.

Use the information in the table to support your answer.

(2)

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(e) Describe how the scientist could extend this investigation to collect more data to either support or reject the hypothesis.

(2)

(Total for Question 3 = 14 marks)
